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Multibody Analysis and Optimisation of a Motorcycle Rear Suspension

Multibody Analysis and Joint-Load Reduction
in MSC Adams Student Edition

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Project overview

An academic university project using MSC Adams Student Edition to model and optimise a motorcycle rear suspension under prescribed wheel motion. The study combines kinematic analysis, joint-reaction evaluation, parameter optimisation and a design-of-experiments study to reduce joint loads while satisfying a spring-force requirement. A subsequent bushing study examines how connection compliance changes force transmission.

Abstract

This report presents the modelling, analysis and optimisation of a simplified motorcycle rear suspension using MSC Adams Student Edition. Starting from a supplied layout containing the main geometric reference points and four rigid bodies, a working multibody model is developed through the definition of joints, a prescribed wheel motion and a spring–damper connection.

The wheel is driven by a sinusoidal vertical input with an amplitude of 100 mm and a frequency of 1 Hz. Kinematic measures are used to examine displacement, velocity and acceleration, while force measures identify the most highly loaded joints. The initial connecting-rod joint reaction reaches approximately 3.67 kN. The peak absolute spring–damper force is approximately 2.98 kN, within the specified interval of 2.50–3.00 kN.

A first optimisation substantially reduces the joint reactions but also lowers the spring–damper force below the required interval. A subsequent design-of-experiments study is therefore used to select a configuration that satisfies the force requirement while retaining a reduction in joint loading. The selected configuration produces a peak connecting-rod joint reaction of approximately 2.64 kN, about 28% below the initial value, and a peak absolute spring–damper force of 2.56 kN.

Finally, the cylindrical joint between the connecting rod and the swingarm is replaced by a compliant bushing. Dynamic simulations examine the resulting changes in transmitted forces and compare the wheel's vertical kinematics with those of the rigid-joint model. A repeated comparison indicates a spring-force peak of approximately 2.0 kN with the bushing, below the specified interval; the rigid-joint design therefore remains the configuration satisfying that requirement. The report documents the modelling choices, simulation settings, optimisation sequence and graphical comparisons used throughout the study.

The work demonstrates multibody modelling in MSC Adams/View, definition of motion and force measures, interpretation of results in Adams/PostProcessor, and parameter-based design evaluation through optimisation and a full-factorial DOE. The emphasis is on relating modelling choices to measurable changes in force transmission.

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Chapter 1

Introduction

1.1 Aim of the project

The aim of this project is to model and analyse a motorcycle rear suspension and to investigate how its geometric and elastic parameters affect the loads transmitted through the mechanism. The study is carried out in MSC Adams Student Edition, using a prescribed motion to provide a consistent excitation for the different configurations.

The starting point is a predefined geometric layout containing the suspension components and their principal reference points. The modelling work consists of defining suitable kinematic connections, applying the wheel motion, introducing a spring–damper element and creating the measures needed to evaluate the response. The analysis then progresses from the initial configuration to an optimised configuration and, finally, to a model incorporating a compliant connection.

1.2 Objectives and sequence of the study

The principal design objective is to reduce the largest joint reactions while maintaining the peak absolute force in the spring–damper system within the required interval of 2500–3000 N. This interval refers to the maximum absolute force during the simulation, rather than to the instantaneous force throughout the cycle.

The work is organised into the following stages:

1. Definition of the joints and verification of the model's degrees of freedom.
2. Application of the sinusoidal input and assessment of the wheel's vertical kinematics.
3. Evaluation of joint reactions and the force developed by the spring–damper system.
4. Minimisation of the main reactions through changes to selected design variables.
5. Adjustment of the design through a design-of-experiments study to satisfy the spring–damper force requirement.
6. Introduction of a bushing and comparison of the compliant and rigid-joint responses.

Adams names are retained for traceability: *beam* denotes the connecting rod and *triangle* the rocker arm. Units are N, mm and s unless stated otherwise.

Chapter 2

System modelling

2.1 Components and geometric layout

The suspension is represented by four moving rigid bodies: the rear wheel, the swingarm, the connecting rod and the rocker arm. Ground represents the chassis reference. The supplied layout includes the principal connection points; the joints are subsequently defined between the appropriate bodies at these locations.

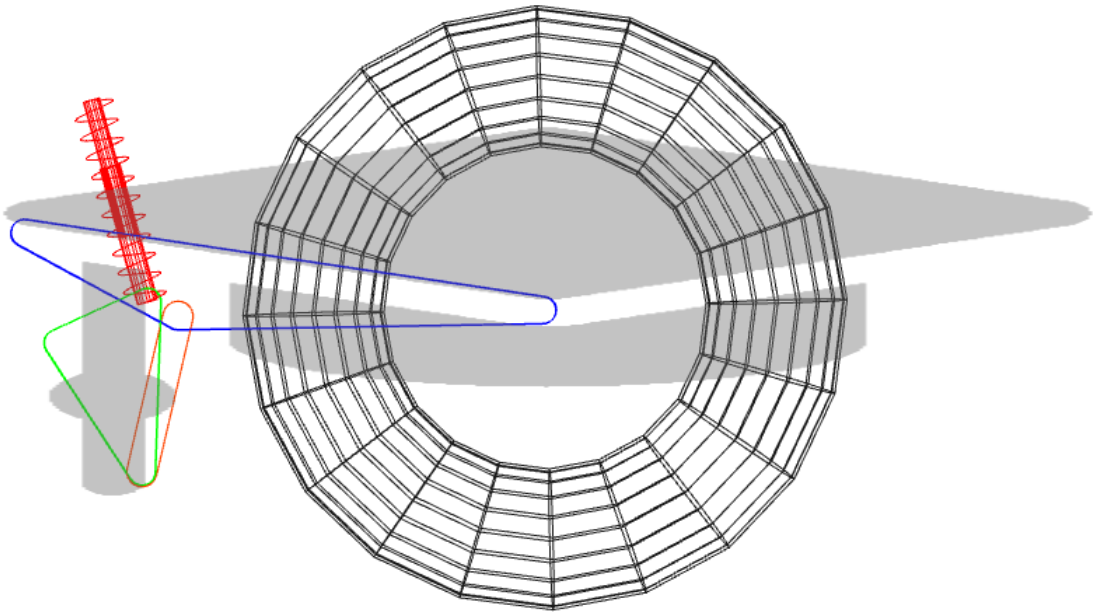


Figure 2.1: Wireframe view of the rear suspension model in MSC Adams: wheel (black), swingarm (blue), rocker arm (green), connecting rod (orange), and spring–damper assembly (red).

Table 2.1: Moving bodies and their roles in the model.

Adams name	Component and role
`wheel`	Rear wheel, whose vertical motion provides the prescribed input.
`swingarm`	Swingarm connecting the wheel to the chassis pivot.
`beam`	Connecting rod transmitting load between the swingarm and rocker arm.
`triangle`	Rocker arm connected to the chassis and spring–damper system.

2.2 Kinematic connections

The joints are chosen to permit the intended motion while avoiding redundant constraints. Their locations follow the geometric reference points of the supplied model.

2.2.1 Swingarm and rocker pivots

A revolute joint connects the swingarm to ground at `P_swar_ch`. It allows the swingarm to rotate about its chassis pivot. A second revolute joint connects the rocker arm to ground at `P_tr_ch`, allowing the rocker to rotate as the connecting rod and spring–damper transmit load through the linkage.

2.2.2 Connecting-rod joints

A spherical joint connects the rocker arm to the connecting rod at `P_beam_t`. This connection permits relative rotation without introducing unnecessary rotational constraints. At the other end, a cylindrical joint connects the rod to the swingarm at `P_swar_b`. Its compatibility with the complete mechanism is checked through the model verification procedure.

2.2.3 Wheel connection and rotation constraint

A revolute joint connects the wheel to the swingarm at `P_whc`. To satisfy the prescribed condition of zero relative wheel rotation, a constant zero motion, named `blocco_rotazione_wheel`, is applied to this joint.

2.3 Degrees of freedom

After the joints and wheel rotation constraint are defined, Model Verify identifies one remaining degree of freedom and no redundant constraint equations. This remaining freedom is controlled by prescribing the vertical component of the wheel motion. Once that input is added, the model has zero remaining degrees of freedom.

Table 2.2: Verification of the rigid-joint model.

Model state	Remaining DOFs	Redundancies
Joints and wheel rotation constraint	1	None
Vertical wheel input added	0	None

The verification establishes that the prescribed-motion model is kinematically determined. The input controls a displacement component of the wheel; the remaining components of its trajectory follow from the linkage geometry and constraints.

2.4 Definition of the prescribed motion

A General Motion is defined between the wheel and ground at `P_whc`. An auxiliary point, `punto_ausiliario1`, has the same X and Z coordinates as this point and a larger Y coordinate. It is used to establish the vertical direction of the motion reference.

The prescribed vertical displacement is

$$\Delta y(t) = A\sin(2\pi f t), \qquad A = 100\,\text{mm}, \quad f = 1\,\text{Hz}. \tag{2.1}$$

The input therefore produces a peak-to-peak excursion of 200 mm and two complete cycles during a 2 s simulation. Here Δy denotes the imposed displacement relative to the initial reference; an absolute-position measure may also contain the initial coordinate offset.

2.5 Spring–damper system

A translational spring–damper, named `molla_triangle_ground`, connects `P_cyl_tr` on the rocker arm to `P_cyl_ch` on ground. The rocker is the action body and ground is the reaction body.

The nominal parameter values follow the prescribed university project settings. The model uses the design variable `DV_K` to link stiffness, damping and preload. Writing its numerical value as p, the relationships are

$$K = p\,\text{N/mm}, \qquad C = \frac{p}{100}\,\text{N s/mm}, \qquad F_0 = p\,\text{N}. \tag{2.2}$$

The nominal value is $p = 75$, with an allowed variation of $\pm 10\%$, giving $67.5 \le p \le 82.5$. Thus changes to this variable modify all three linked spring–damper properties together.

Table 2.3: Nominal spring–damper properties.

Property	Value	Unit
Stiffness K	75	N/mm
Damping coefficient C	0.75	N s/mm
Preload F_0	75	N

The coefficient fields are linked to the design variable in Adams so that each parameter change is propagated consistently during the subsequent design studies.

Chapter 3

Kinematic simulation

3.1 Simulation settings and measures

The initial kinematic analysis checks the vertical displacement, velocity and acceleration of the rear wheel under the input defined in Equation (2.1). The simulation is performed in Kinematics mode for 2 s with 200 output steps, corresponding to a nominal output spacing of 0.01 s, or 100 steps per second.

Three measures are defined at the wheel centre of mass, along the Y direction:

- `tra_Y_wheel`: vertical displacement or position output, in mm;
- `vel_Y_wheel`: vertical velocity, in mm/s;
- `acc_Y_wheel`: vertical acceleration, in mm/s^2.

The analytical derivatives of the motion law provide a reference for interpreting these outputs.

3.2 Vertical displacement

The displacement is sinusoidal and completes two cycles over the simulated interval. Its amplitude relative to the initial reference is 100 mm, giving a total vertical excursion of 200 mm. Any constant offset in the plotted ordinate reflects the selected position reference and does not change this excursion.

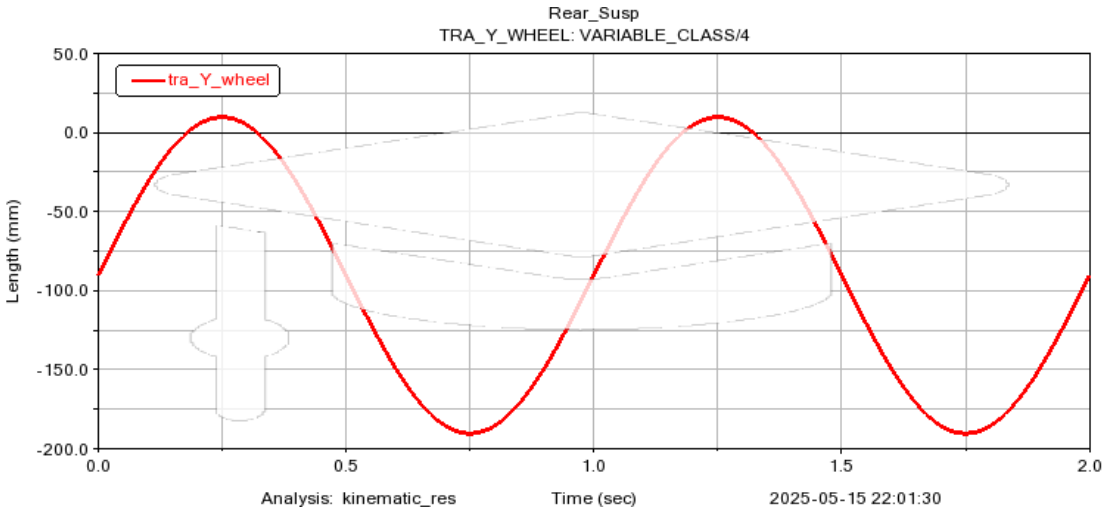


Figure 3.1: Vertical wheel displacement in the initial configuration.

3.3 Vertical velocity and acceleration

Differentiating the prescribed displacement gives

$$v_y(t) = 2\pi f A \cos(2\pi f t). \tag{3.1}$$

The velocity amplitude is approximately 628 mm/s. Positive peaks occur at $t = 0$, 1 and 2 s; negative peaks occur at $t = 0.5$ and 1.5 s. The velocity is zero at the displacement extrema, when the wheel reverses direction.

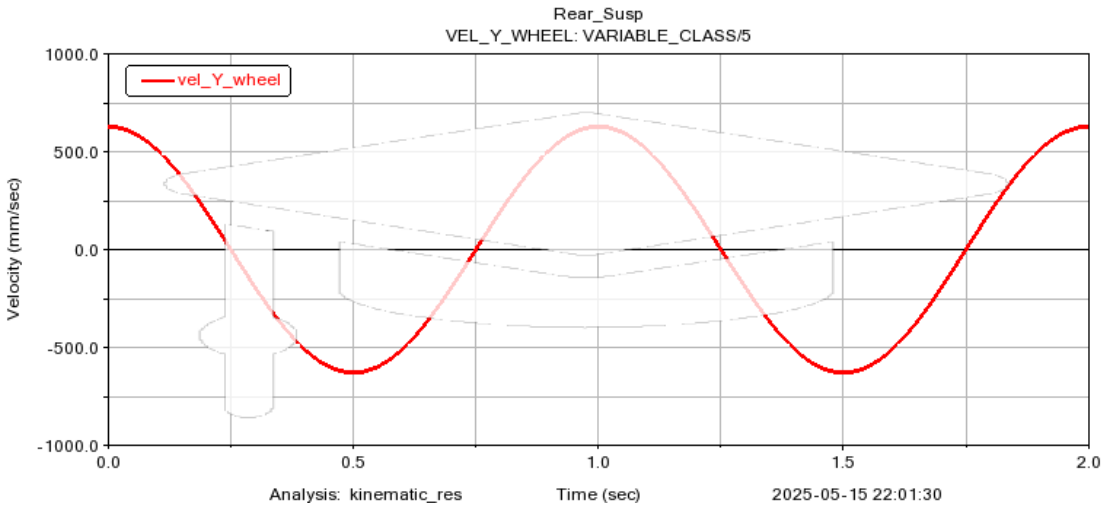


Figure 3.2: Vertical wheel velocity under the prescribed sinusoidal input.

The acceleration follows from a second differentiation:

$$a_y(t) = -(2\pi f)^2 A \sin(2\pi f t). \tag{3.2}$$

Its amplitude is approximately 3.95×10^3 mm/s^2. Acceleration is zero at the velocity extrema and reaches its greatest magnitude when velocity is zero. The plotted phase relationships agree with these expressions.

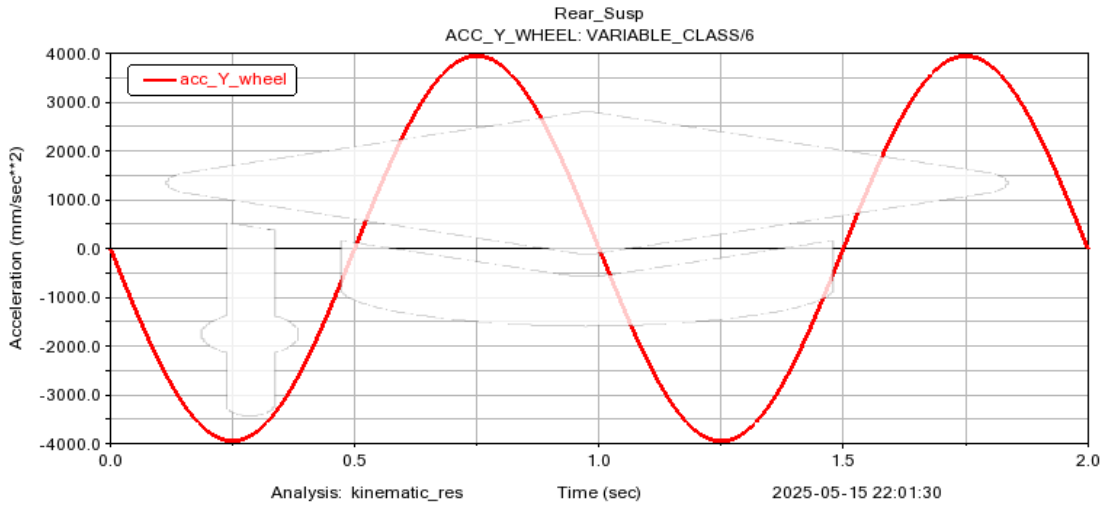


Figure 3.3: Vertical wheel acceleration in the initial configuration.

Chapter 4

Constraint reactions and spring force

4.1 Measures and evaluation criteria

The force analysis identifies the most highly loaded joints and evaluates the spring–damper force. A separate measure is created for each joint, using the magnitude of the resultant force, denoted MAG in Adams. The same 2 s prescribed-motion analysis and 200 output steps are used as in the kinematic study.

Table 4.1: Force measures used for the rigid-joint model.

Measure name	Connection or quantity
`reazione_swingarm_ground`	Swingarm–chassis reaction
`reazione_triangle_ground`	Rocker–chassis reaction
`reazione_swingarm_wheel`	Swingarm–wheel reaction
`reazione_beam_swingarm`	Connecting rod–swingarm reaction
`reazione_beam_triangle`	Connecting rod–rocker reaction
`forza_molla`	Signed spring–damper force

For a joint-force vector $\mathbf{R}(t)$, the plotted magnitude is

$$R(t) = \sqrt{R_x(t)^2 + R_y(t)^2 + R_z(t)^2}. \tag{4.1}$$

These plots are non-negative. By contrast, the spring–damper force retains its sign, so its positive and negative peaks distinguish the two loading directions. The force requirement is assessed using the maximum absolute value over the full simulation interval.

4.2 Assessment of the initial configuration

The following figures show the outputs obtained in the Adams PostProcessor. Their relative amplitudes identify where the subsequent optimisation should focus. The chassis pivots carry intermediate reaction levels, whereas the two connecting-rod joints exhibit the largest force magnitudes.

4.2.1 Reactions at the chassis pivots

The swingarm–ground reaction is periodic, with a maximum slightly above 2500 N. The rocker–ground reaction has a similar periodic pattern and a maximum slightly below 2500 N. These curves show how the prescribed wheel motion loads the two chassis connections during each cycle.

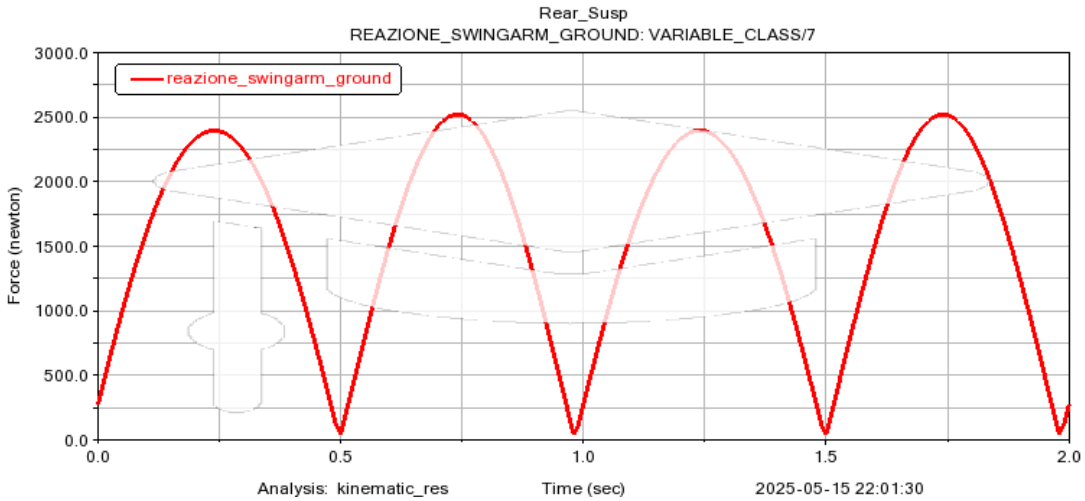


Figure 4.1: Initial reaction-force magnitude at the swingarm–ground joint.

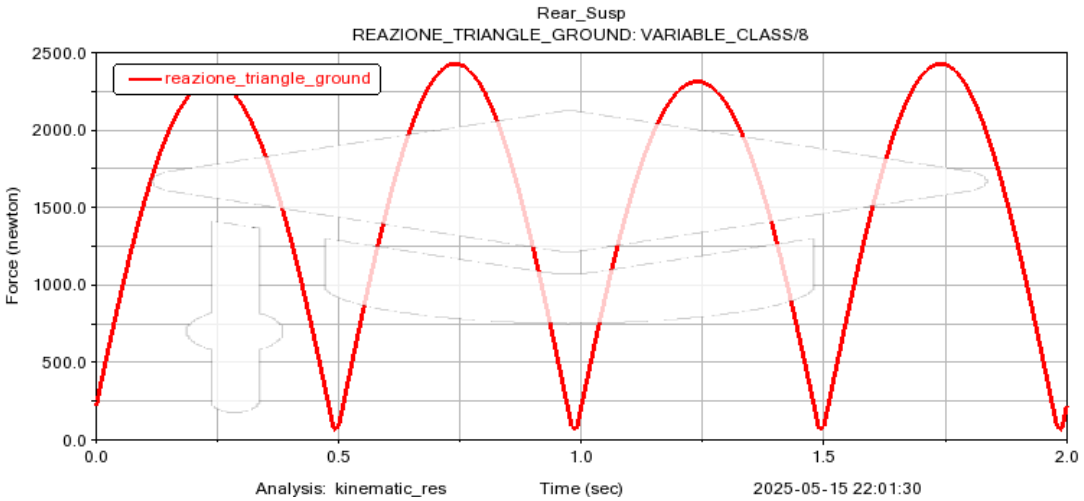


Figure 4.2: Initial reaction-force magnitude at the rocker–ground joint.

4.2.2 Wheel and connecting-rod reactions

The swingarm–wheel reaction reaches approximately 1200 N, making it the least highly loaded of the joints considered. At the connecting rod–rocker joint, the reaction reaches approximately 3700 N, substantially above the chassis-pivot and wheel-joint values.

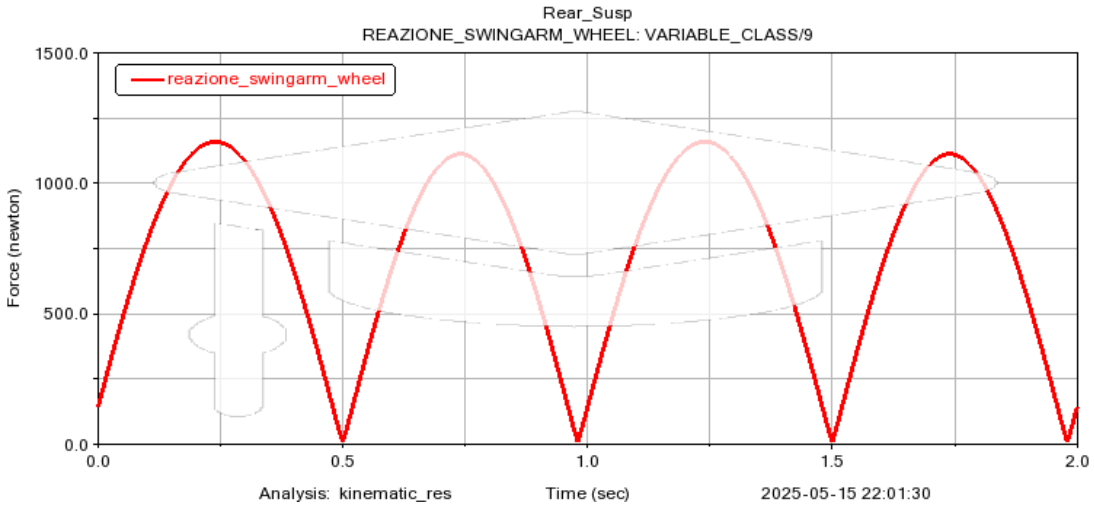


Figure 4.3: Initial reaction-force magnitude at the swingarm–wheel joint.

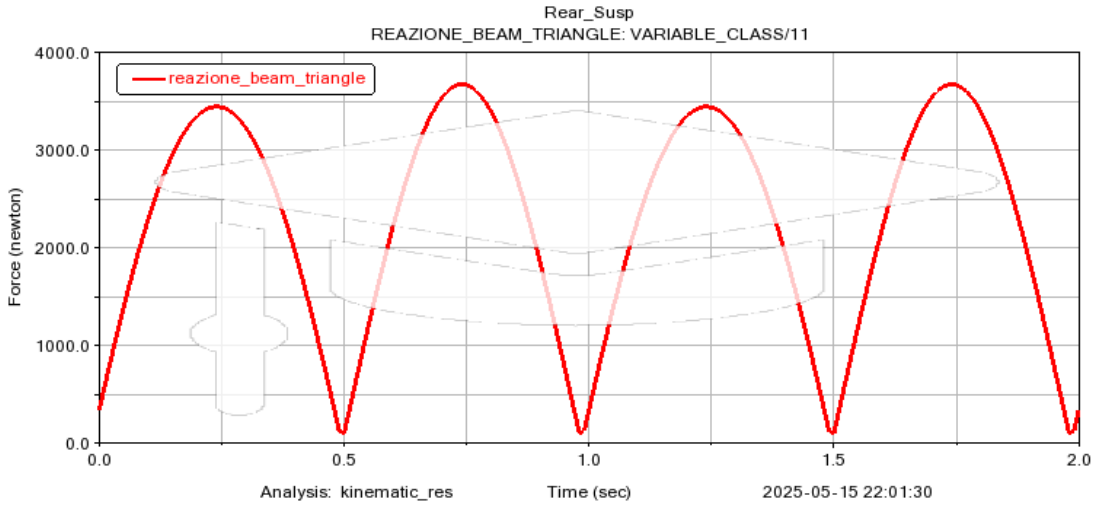


Figure 4.4: Initial reaction-force magnitude at the connecting rod–rocker joint.

The connecting rod–swingarm reaction also reaches approximately 3700 N. The two connecting-rod reaction magnitudes exhibit closely matching trends in this model. These connections are therefore selected as the focus of the load-reduction study, with the beam–swingarm reaction used as the optimisation objective.

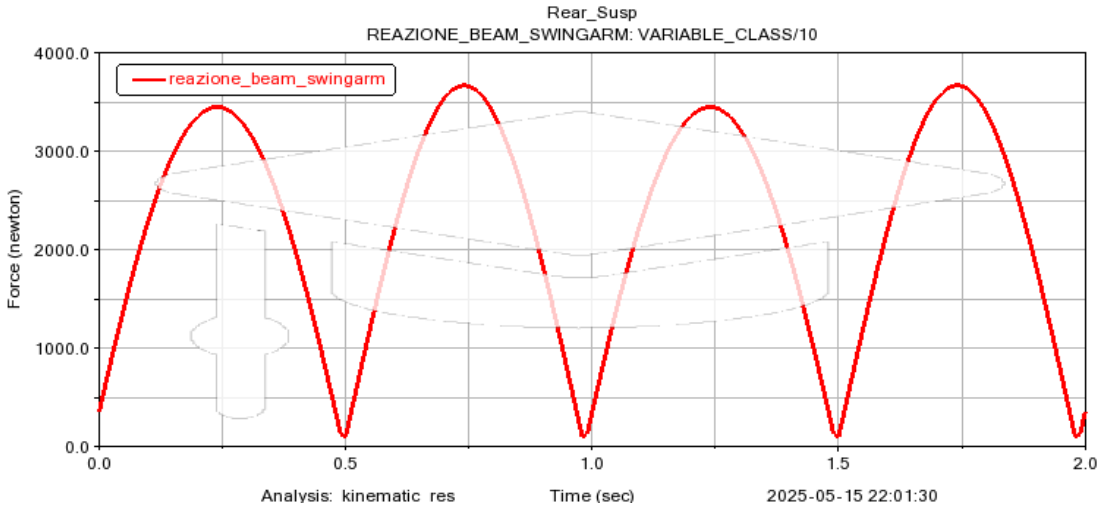


Figure 4.5: Initial reaction-force magnitude at the connecting rod–swingarm joint.

4.2.3 Spring–damper force

The spring–damper force alternates in sign over the cycle and has a peak absolute value just below 3000 N. To evaluate it directly, the objective `max_forza_molla` is defined as

$$F_{\text{peak}} = \max_{0 \le t \le 2\,\text{s}} |F_{\text{sd}}(t)| \,. \tag{4.2}$$

Evaluation of the initial simulation gives approximately 2982 N, which satisfies the specified 2500–3000 N interval.

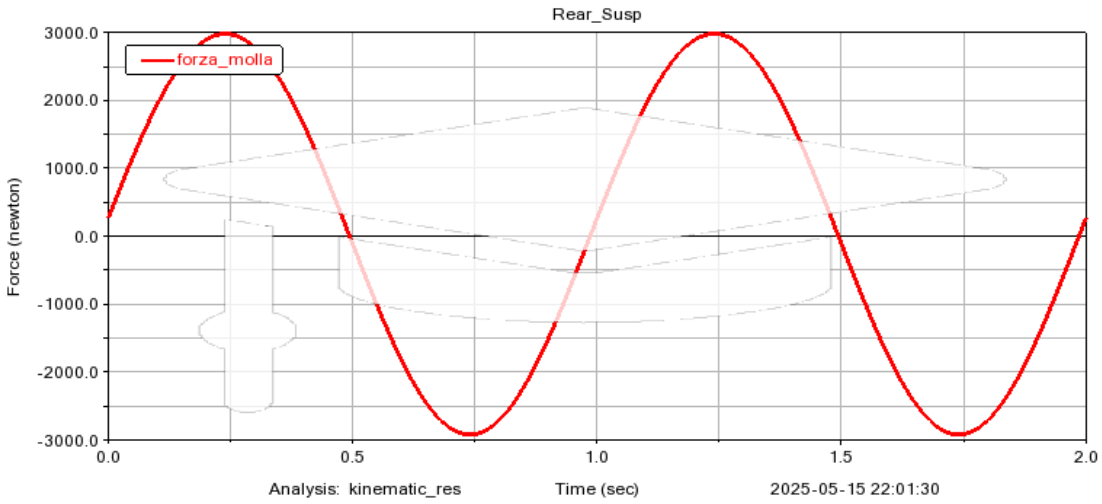


Figure 4.6: Signed spring–damper force in the initial configuration.

Chapter 5

Minimisation of the main reactions

5.1 Design objective

The initial force analysis identifies the connecting-rod joints as the most highly loaded connections. The beam–swingarm reaction is selected for minimisation, while the response at the beam–triangle connection is also monitored throughout the study.

The design objective `max_reazione_beam_swingarm` evaluates the maximum reaction magnitude over the simulation:

$$J_R(\mathbf{p}) = \max_{0 \le t \le 2\,\mathrm{s}} R_{\mathrm{beam-swingarm}}(t;\mathbf{p}), \tag{5.1}$$

where $\mathbf{p}$ denotes the selected design variables. Evaluation of the initial configuration gives $J_R \approx 3668.3$ N, or approximately 3670 N when rounded.

5.2 First optimisation

5.2.1 Design variables and solver settings

The Design Evaluation Tool is configured to minimise Equation (5.1). Seven variables are selected: `DV_K`, `x_mono`, `y_mono`, `x_triangle`, `y_triangle`, `delta_tx` and `delta_ty`. Their values and intervals are listed in Appendix A.

The geometric variables alter the locations of the spring–damper and linkage reference points. The variable `DV_K` changes the linked spring–damper properties according to Equation (2.2).

Table 5.1: Settings used for the first optimisation.

Setting	Value
Objective	Minimise maximum beam–swingarm reaction
Algorithm	MSCADS-MMFD
Tolerance	10^{-3}
Maximum iterations	100
Differencing	Centred

5.2.2 Parameter changes and optimisation history

The optimisation record contains the initial configuration and two subsequent updates. The reaction objective falls from 3668.3 N to 1884.1 N and then to 1750.8 N. The corresponding changes are concentrated in `DV_K`, `y_triangle` and `delta_tx`; the other selected variables retain their initial values in this run.

Table 5.2: Recorded optimisation history. Geometric coordinates are in mm.

Record	J_R [N]	`DV_K`	`y_triangle`	`delta_tx`
0	3668.3	75.0	-160	10.000
1	1884.1	67.5	-110	22.191
2	1750.8	67.5	-110	30.000

The final changes are a 10% reduction in `DV_K`, a 50 mm increase in `y_triangle` and a 20 mm increase in `delta_tx`. Together, they substantially reduce the reaction peaks. Figure 5.1 compares the initial and first-optimisation histories at the selected joint.

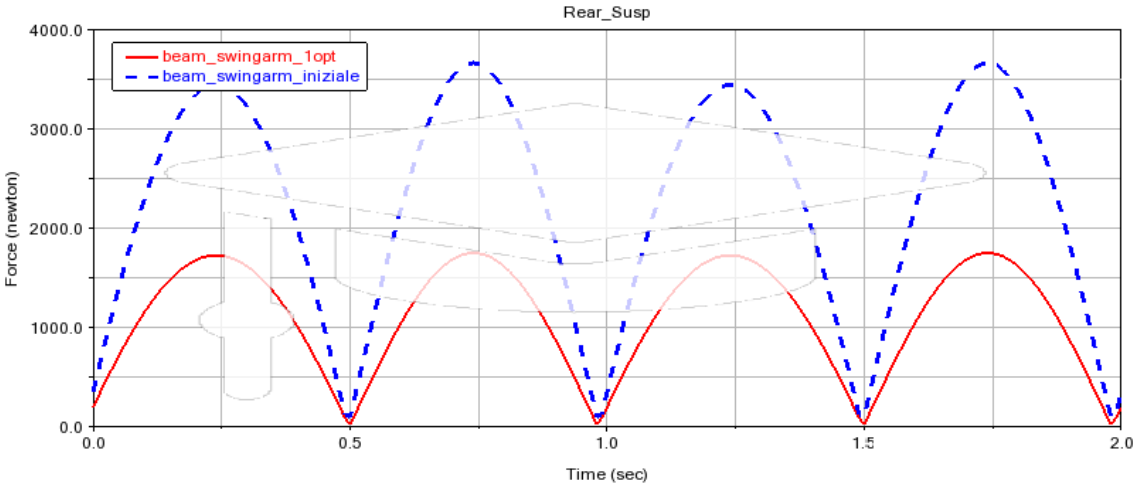


Figure 5.1: Beam–swingarm reaction before and after the first optimisation. Blue dashed: initial configuration; red solid: first optimised configuration.

Changes to the spring parameter affect stiffness, damping and preload together, rather than stiffness alone. The joint-reaction objective alone is not sufficient to select the final design. The spring–damper force must also remain within its required interval, as examined below.

5.2.3 Effect on the other monitored forces

The beam–triangle reaction follows the same reduction in the first optimised configuration. However, the spring–damper force also decreases substantially. Its peak absolute value falls below the lower bound of the prescribed interval.

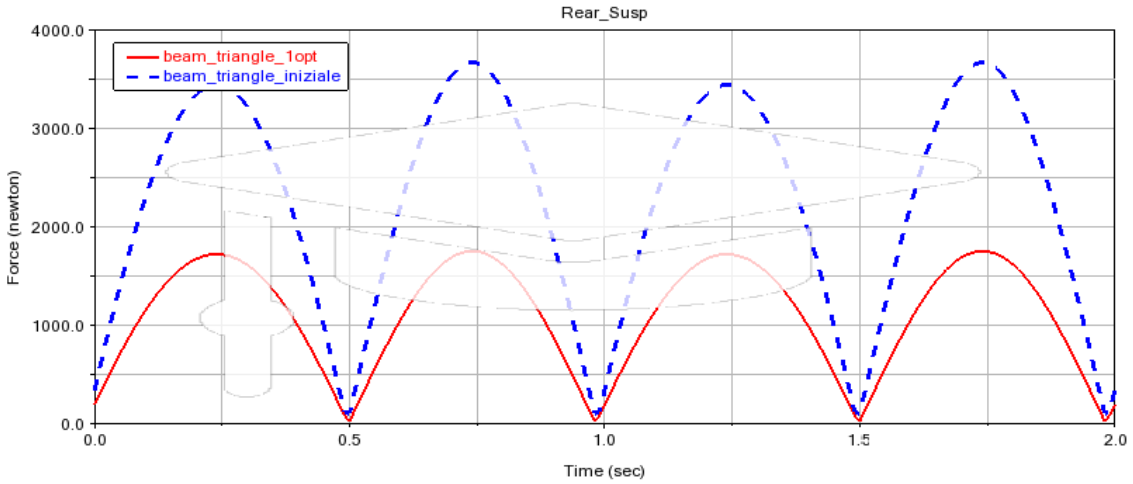


Figure 5.2: Beam–triangle reaction before and after the first optimisation. Blue dashed: initial; red solid: first optimised configuration.

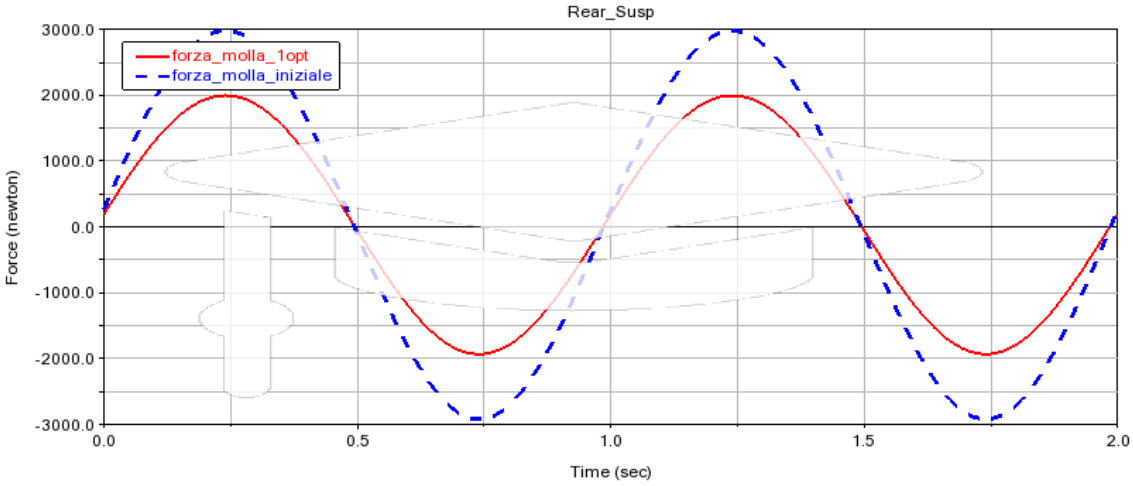


Figure 5.3: Spring–damper force before and after the first optimisation. The reduction in force amplitude leads to a peak below the required interval.

5.3 Second design stage

5.3.1 Reassessment of the spring–damper force

After the first optimisation, `DV_K` is at its lower allowed value of 67.5. Evaluation of `max_forza_molla` gives approximately 1997 N, below the 2500 N minimum required for the peak force.

As an initial adjustment, `DV_K` is restored to its nominal value of 75 while retaining the modified geometry. The resulting simulation, `post_prima_opt_sim`, gives a peak force of approximately 2220 N. This is an increase, but it remains below the required range.

Table 5.3: Spring-force assessment during the intermediate design stages.

Configuration	`DV_K`	Peak force [N]
Initial model	75	≈ 2982
First optimisation	67.5	≈ 1997
Modified geometry, nominal spring parameter	75	≈ 2220

5.3.2 Design of experiments

To recover a peak force within 2500–3000 N, a design-of-experiments study is performed on five variables: `DV_K`, `x_mono`, `y_mono`, `x_triangle` and `delta_ty`. The previously modified values of `y_triangle` and `delta_tx` are retained.

The DOE uses a full-factorial design with three levels per factor. With five factors, the configured number of runs is $3^5 = 243$. The objective evaluated in the DOE is the maximum absolute spring–damper force. This stage explores combinations that can satisfy the force requirement while preserving the benefit of the earlier geometry changes.

A full-factorial design evaluates every combination of the prescribed factor levels. Unlike varying one parameter at a time, it therefore samples simultaneous changes in spring properties and attachment-point coordinates. This is relevant to a linkage in which the effect of a spring parameter depends on the geometry through which its force is transmitted.

Here the DOE serves as a selection step based on the spring-force requirement. It does not establish a global minimum of the joint-reaction objective over all feasible designs. After a candidate is selected, both the spring force and the joint reactions must be reassessed to determine the compromise actually obtained.

5.3.3 Selected DOE configuration

The DOE summary identifies a maximum spring-force objective of 2562.81 N at trial 225 and a minimum of 1938.33 N at trial 4. Trial 225 is selected because its peak force lies within the required interval.

Table 5.4: Parameter values of the selected DOE configuration.

Variable	Value	Unit
`DV_K`	82.5	–
`x_mono`	70	mm
`y_mono`	130	mm
`x_triangle`	30	mm
`y_triangle`	-110	mm
`delta_tx`	30	mm
`delta_ty`	15	mm

The selected values are applied to the model, and a verification simulation is performed using `second_opt_sim`. The spring–damper force reaches a maximum absolute value of 2562.8 N. Figure 5.4 compares this force history with that obtained after the first optimisation.

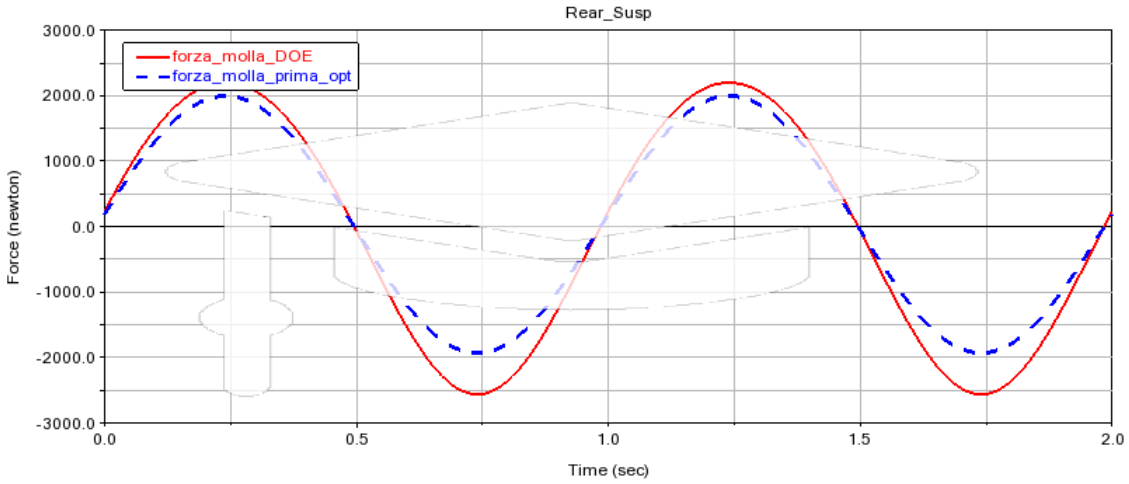


Figure 5.4: Spring–damper force after the DOE selection compared with the first optimised configuration. Red solid: selected DOE design; blue dashed: first optimisation.

The positive and negative peaks are no longer symmetric. This is consistent with the changed linkage geometry and the resulting difference in mechanical leverage during the two phases of motion. Under the sign convention used in the study, the largest negative force corresponds to compression.

5.3.4 Connecting-rod reactions after DOE selection

Restoring the spring-force peak to the required range increases the beam–swingarm reaction relative to the first optimisation. Its peak rises from approximately 1750 N to 2643.4 N, but remains about 1000 N below the initial value. The beam–triangle reaction shows the corresponding change.

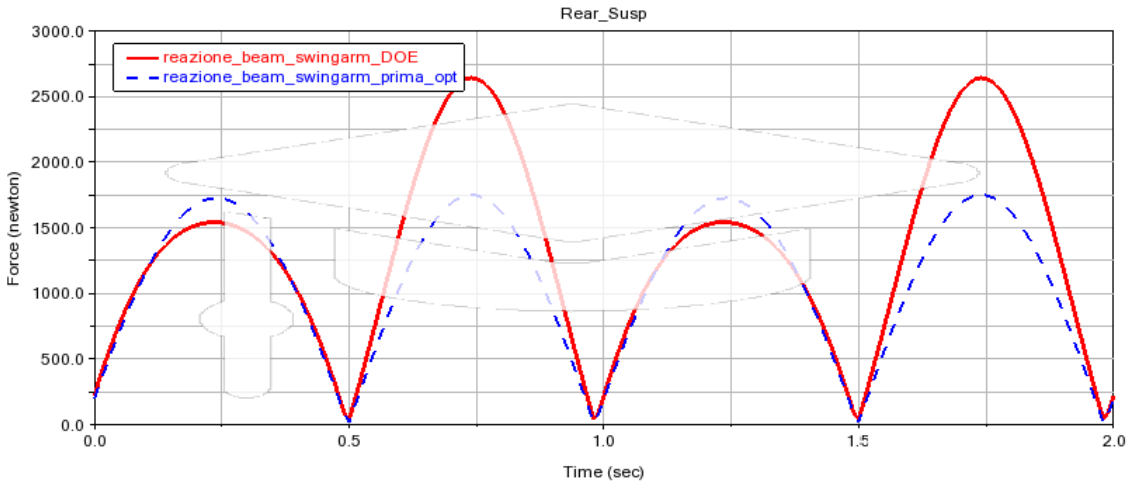


Figure 5.5: Beam–swingarm reaction after DOE selection compared with the first optimisation.

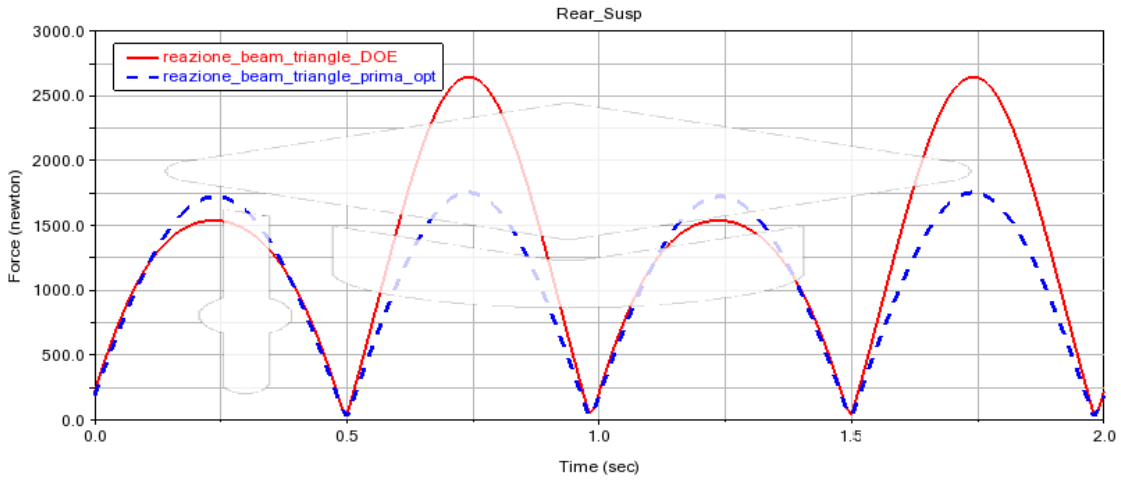


Figure 5.6: Beam–triangle reaction after DOE selection compared with the first optimisation.

5.4 Final comparison with the initial model

5.4.1 Chassis-pivot reactions

The final rigid-joint configuration is compared with the initial model using the same prescribed input. In the following plots, the final response is shown by the red solid curve and the initial response by the blue dashed curve.

The swingarm–ground peak decreases from approximately 2500 N to 1800 N, a reduction of about 28%. The rocker–ground peak decreases from approximately 2500 N to 1750 N, corresponding to about 30%.

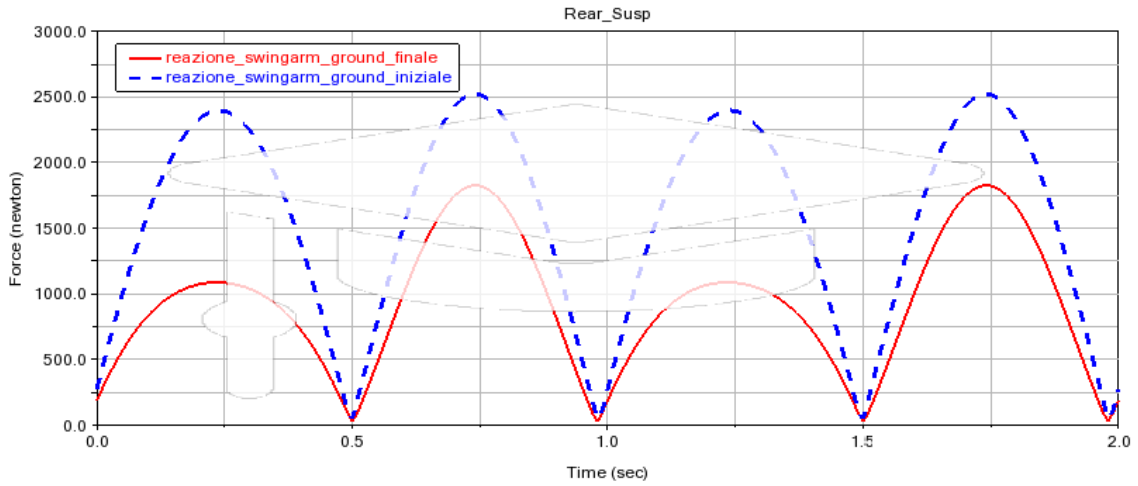


Figure 5.7: Final and initial swingarm–ground reaction-force magnitudes.

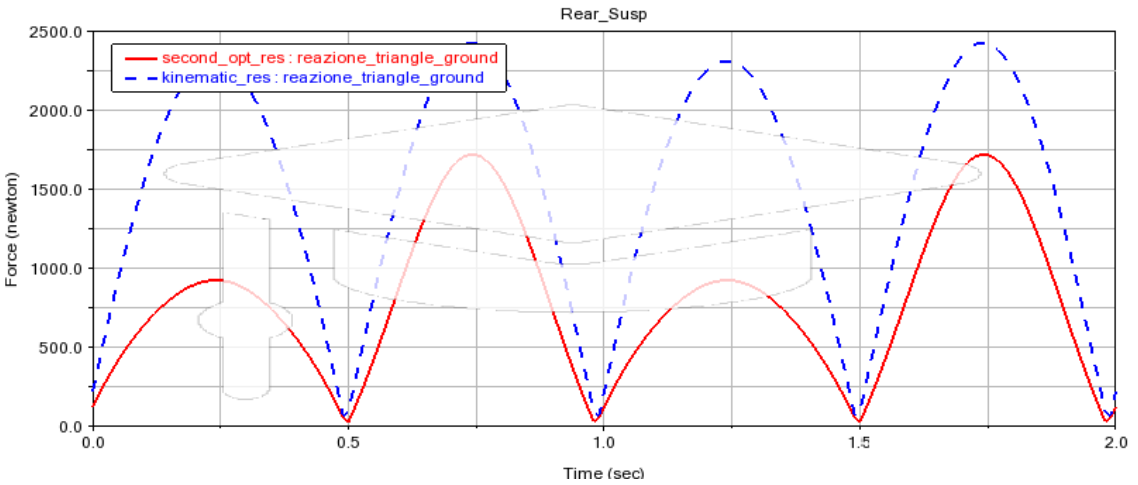


Figure 5.8: Final and initial rocker–ground reaction-force magnitudes.

5.4.2 Wheel and connecting-rod reactions

The swingarm–wheel reaction shows a clear reduction in peak magnitude. Its reduction is assessed graphically here, without assigning a percentage to the approximate peak readings. The connecting rod–rocker peak falls from approximately 3670 N to 2640 N, a reduction of about 28%.

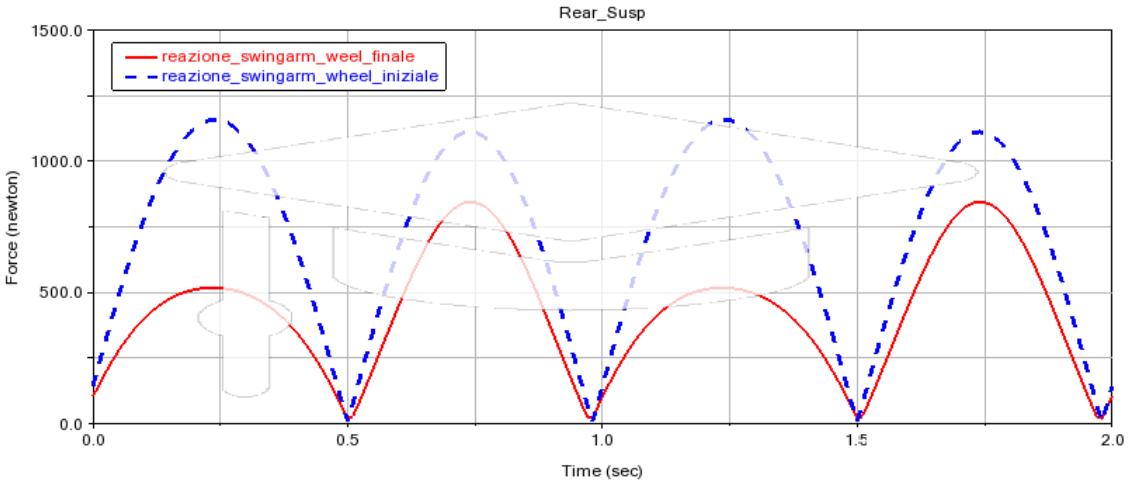


Figure 5.9: Final and initial swingarm–wheel reaction-force magnitudes.

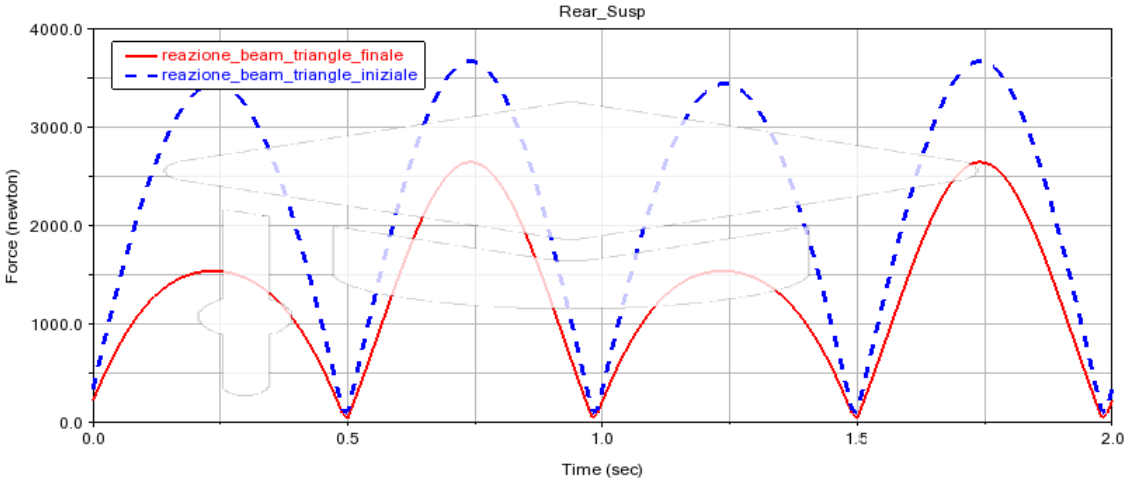


Figure 5.10: Final and initial connecting rod–rocker reaction-force magnitudes.

The beam–swingarm connection exhibits the same overall reduction, with a final peak of 2643.4 N. The maximum absolute spring–damper force decreases from approximately 2982 N to 2562.8 N, a reduction of about 14%, while satisfying the 2500–3000 N requirement.

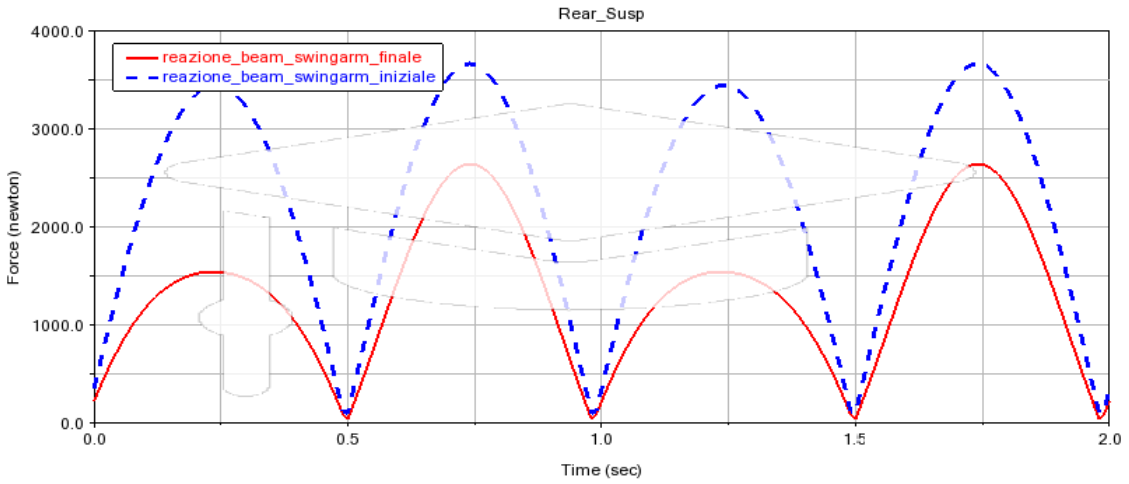


Figure 5.11: Final and initial beam–swingarm reaction-force magnitudes.

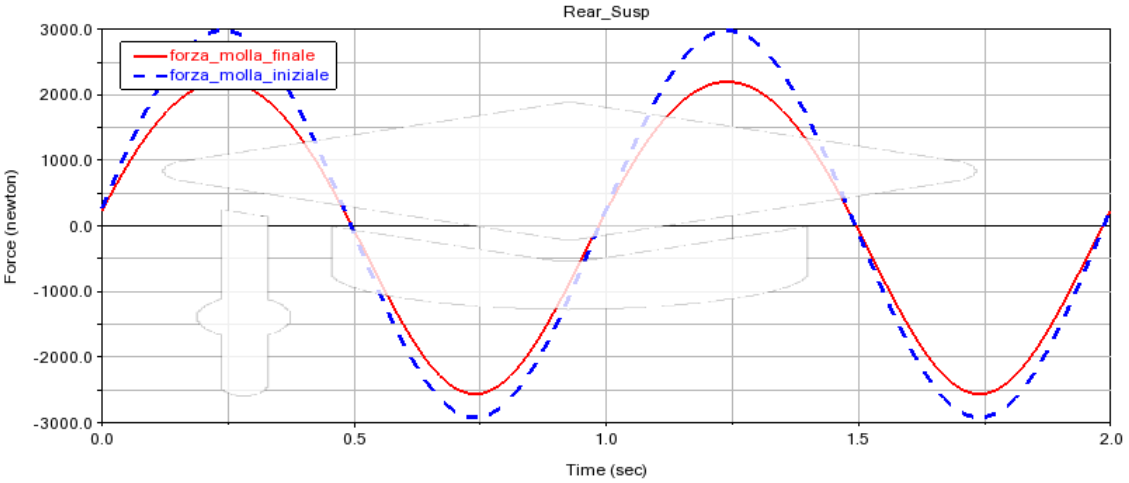


Figure 5.12: Final and initial spring–damper forces. The final peak absolute force remains within the required interval.

5.4.3 Summary of the rigid-joint optimisation

Table 5.5 collects the main comparisons. Values read approximately from the plotted histories are distinguished from the evaluated design-objective values.

Table 5.5: Initial and final results for the rigid-joint model.

Peak force	Initial [N]	Final [N]	Reduction
Swingarm–ground	≈ 2500	≈ 1800	≈ 28%
Rocker–ground	≈ 2500	≈ 1750	≈ 30%
Swingarm–wheel	Lower peak; see Figure 5.9		
Rod–rocker	≈ 3670	≈ 2640	≈ 28%
Rod–swingarm	3668.3	2643.4	≈ 28%
Spring–damper	≈ 2982	2562.8	≈ 14%

The selected configuration achieves the intended compromise: the largest joint reactions are reduced, while the spring–damper peak remains in the specified interval. The first optimisation alone gives smaller joint reactions, but fails the force requirement. The second stage therefore plays an essential role in selecting the final configuration.

The joint-reaction objective and the spring-force requirement should be interpreted together. Minimising joint loading does not automatically provide an acceptable spring–damper response, since changes in geometry and elastic parameters affect the mechanical advantage and force transmission throughout the linkage. The sequence of optimisation, force reassessment and DOE selection makes this interaction explicit.

For the selected design, the spring-force peak is approximately 63 N above the lower bound and 437 N below the upper bound. These differences describe the position of the nominal simulation result within the specified interval; they are not margins established against manufacturing variability or changes in operating conditions. No such variability study is included in the present analysis.

The reported reaction reductions concern force magnitudes at the model connections. They do not directly quantify stresses, fatigue life or bearing durability, which would require component geometry, material data and additional load cases. Nevertheless, they provide a consistent basis for comparing the configurations and identifying the load paths affected by the design changes.

The final rigid-joint configuration is then used as the reference for the compliant-joint study in Chapter 6. This preserves the selected geometric parameters while examining the effect of replacing one ideal connection with an elastic and dissipative element.

Chapter 6

Introduction of joint compliance

6.1 Purpose and location of the bushing

The preceding analyses use ideal joints. In the next stage, a compliant connection is introduced between the connecting rod and swingarm to investigate its effect on the forces transmitted through the suspension.

This location is selected because the beam–swingarm joint is among the most highly loaded connections in the rigid-joint model. The cylindrical joint is deactivated and a bushing is inserted between the same bodies at `P_swar_b`. The connecting rod remains a rigid body; compliance is introduced at the connection itself.

The bushing is named `cedevolezza_beam_swingarm`. Its action body is `beam` and its reaction body is `swingarm`. The associated markers are `beam.MARKER_46` and `swingarm.MARKER_47`.

6.2 Reference loads for parameter selection

The bushing coefficients are selected with reference to the force and moment components obtained from the final rigid-joint simulation. Examining the components separately helps distinguish the loading directions and the relative importance of translational and rotational resistance.

The largest absolute Y force is approximately 2500 N, while the largest absolute X force is approximately 1100 N. The reaction moments about X and Y are very small in the recorded results. The corresponding plots are presented in Figures 6.1–6.4.

These loads provide the basis for assigning different stiffness and damping coefficients along the local directions. The resulting bushing allows relative deformation while transmitting forces between the two connected bodies.

6.2.1 Translational force components

The Y component has the larger force amplitude and reaches its greatest absolute value during the negative part of the cycle. The X component is smaller, with a negative peak of approximately 1100 N. Signed components are shown here, rather than the non-negative force magnitudes used in the main reaction comparisons.

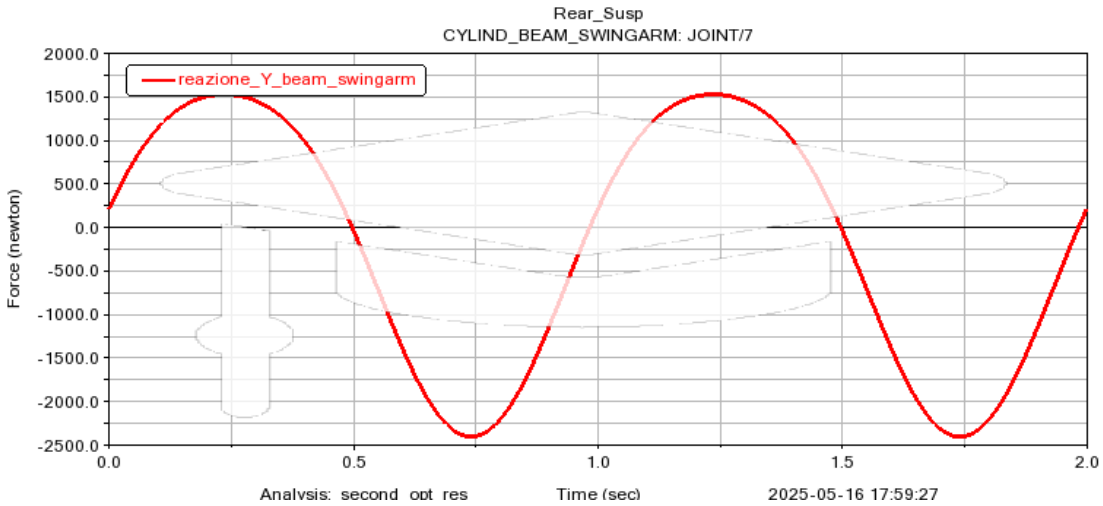


Figure 6.1: Y component of the beam–swingarm reaction in the optimised rigid-joint model.

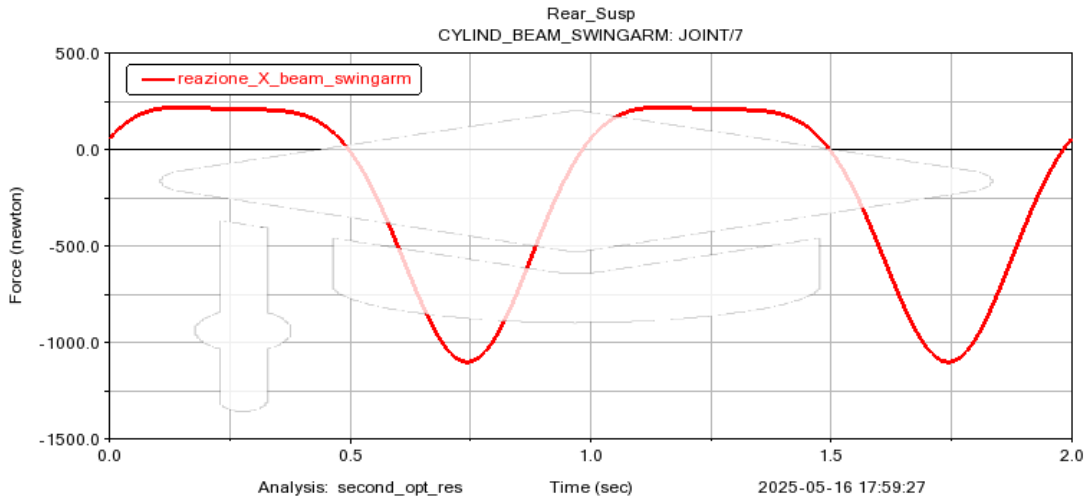


Figure 6.2: X component of the beam–swingarm reaction in the optimised rigid-joint model.

6.2.2 Reaction-moment components

The plotted moments about Y and X are very small. The scientific notation on the ordinate axes is important when interpreting their magnitude: their graphical variation should not be confused with a large transmitted torque.

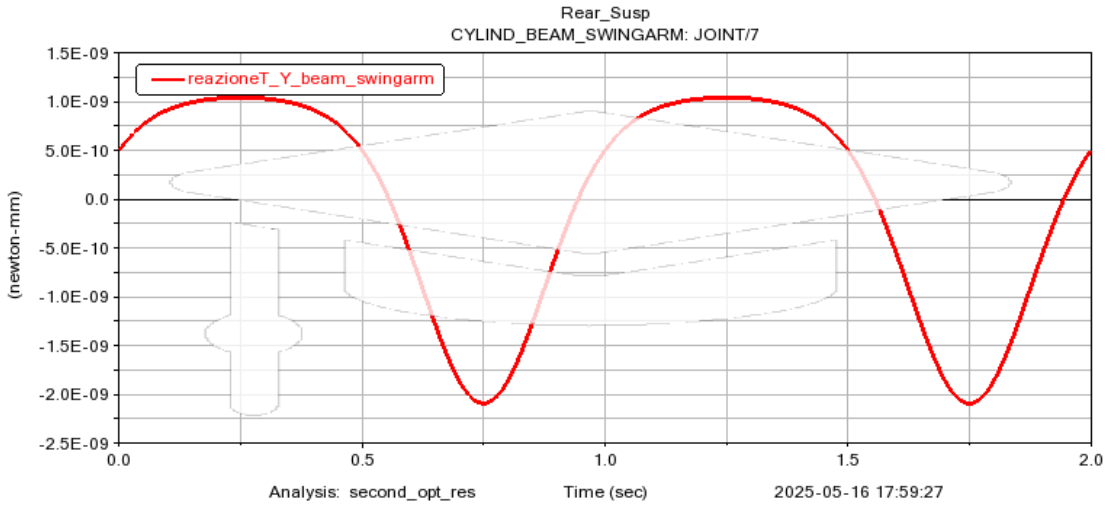


Figure 6.3: Reaction moment about Y at the beam–swingarm joint before introducing compliance.

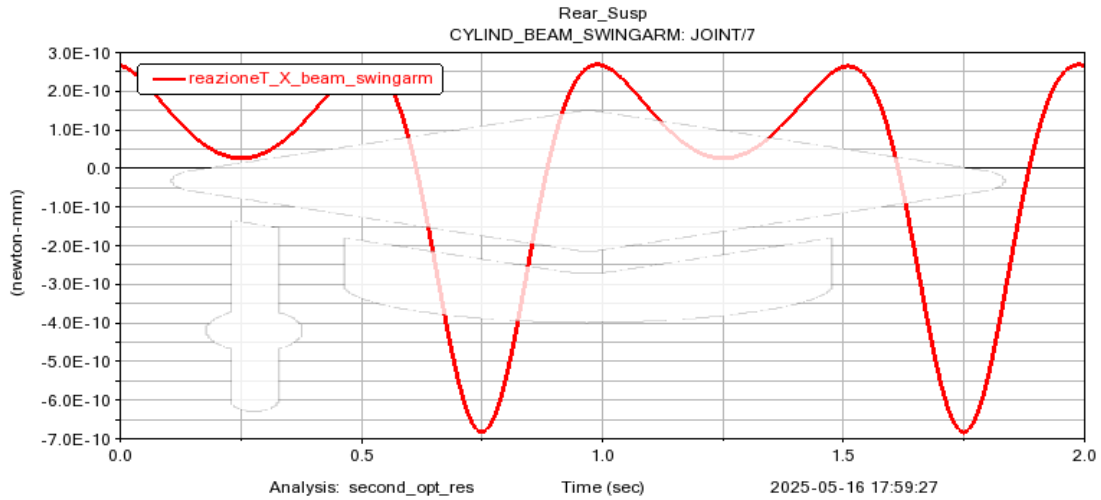


Figure 6.4: Reaction moment about X at the beam–swingarm joint before introducing compliance.

6.3 Bushing properties and model verification

The translational and rotational coefficients are listed in Table 6.1. All force and torque preloads are zero. The coefficients act along the bushing's local X, Y and Z directions.

Table 6.1: Bushing coefficients. Rotational units follow the Adams model settings.

Property	X	Y	Z	Unit
Translational stiffness	100	300	0	N/mm
Translational damping	1	3	0	N s/mm
Rotational stiffness	10^4	10^4	0	N mm/deg
Rotational damping	10	10	0	N mm s/deg

Translation along Z and rotation about Z have zero stiffness and damping, consistent with the free directions of the original cylindrical connection. The other relative motions are resisted by the bushing's elastic and damping forces, rather than prevented by ideal kinematic constraints.

With the cylindrical joint deactivated and the bushing active, Model Verify reports four remaining degrees of freedom and no redundant constraint equations. With the cylindrical joint restored and the bushing inactive, the prescribed-motion rigid model has zero remaining degrees of freedom. The additional freedoms in the compliant model are governed by the dynamic response of the connected bodies.

6.4 Dynamic simulation settings

The compliant model is simulated in Dynamics mode, retaining the prescribed vertical wheel input. The simulation duration remains 2 s, but the number of output steps is increased from 200 to 500. This gives a nominal output spacing of 0.004 s, or 250 steps per second, providing a finer time resolution for the compliance comparison.

Table 6.2: Simulation settings for the two connection models.

Setting	Rigid joints	With bushing
Analysis mode	Kinematics	Dynamics
Duration	2 s	2 s
Output steps	200	500
Nominal output interval	0.01 s	0.004 s

6.5 Force comparison

6.5.1 Connecting-rod connections

A dedicated measure, `bushing_beam_swingarm`, evaluates the magnitude of the force transmitted through the bushing. The following comparisons use a red solid curve for the compliant model and a blue dashed curve for the reference response. At the connecting-rod joints, the reference is the optimised rigid-joint configuration.

The plotted peak forces decrease after the bushing is introduced. The response at the connecting rod–rocker connection shows a corresponding change, reflecting the altered load transmission through the linkage.

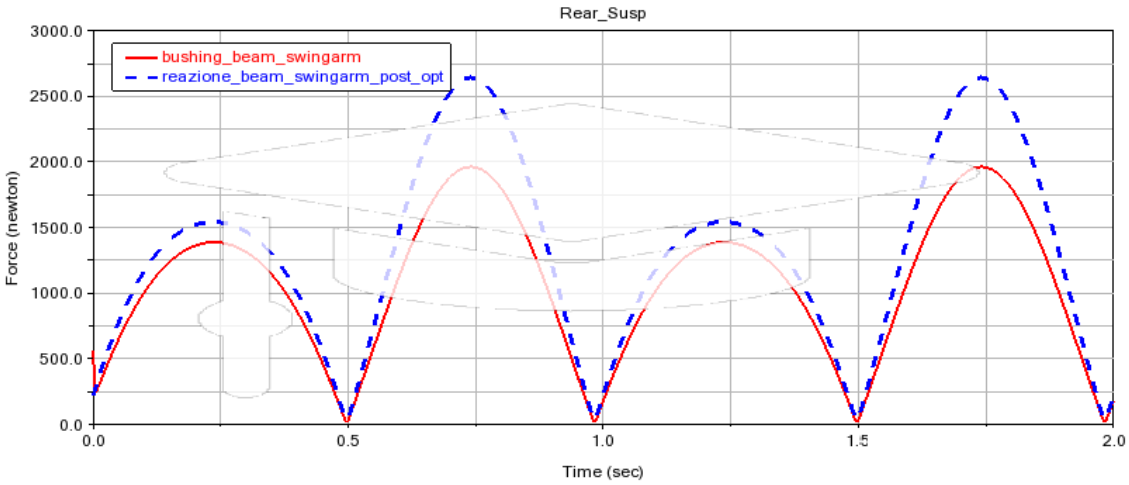


Figure 6.5: Force transmitted between the connecting rod and swingarm: bushing model versus optimised rigid-joint model.

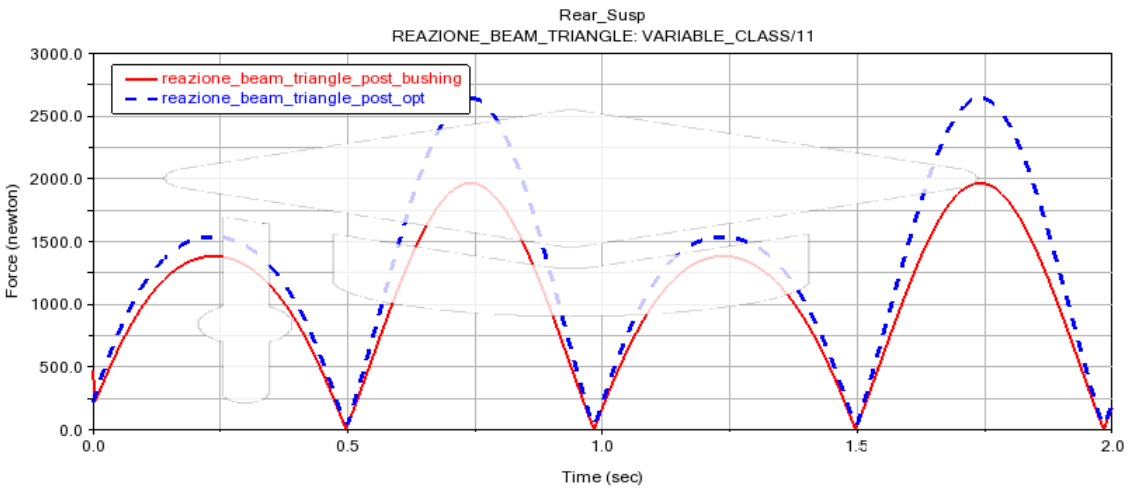


Figure 6.6: Connecting rod–rocker reaction: compliant versus optimised rigid-joint model.

6.5.2 Chassis-pivot reactions

The effect of the compliant connection also appears at the chassis pivots. Both the swingarm–ground and rocker–ground comparisons show reduced peaks relative to the optimised rigid-joint model. Their responses remain periodic under the unchanged sinusoidal excitation.

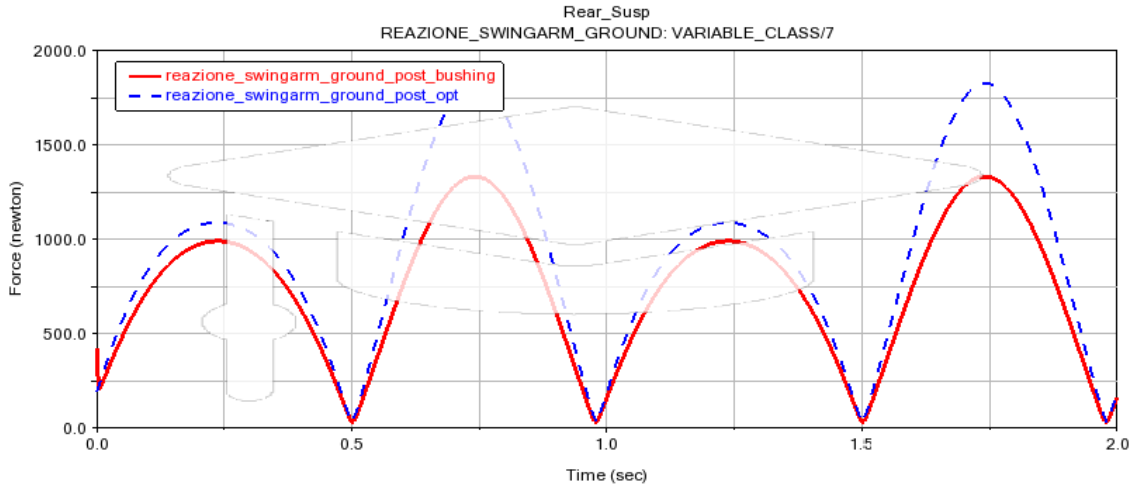


Figure 6.7: Swingarm–ground reaction with and without the compliant connection.

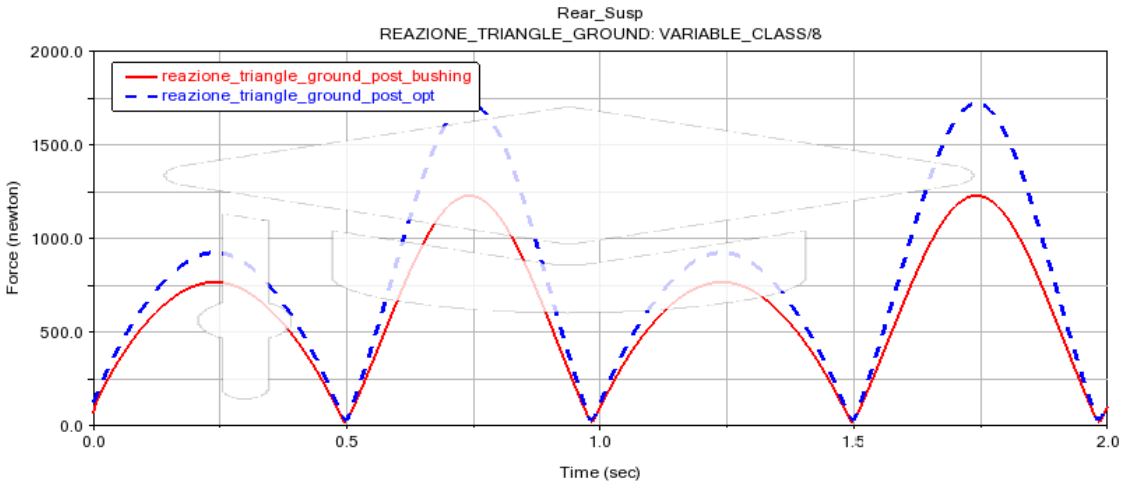


Figure 6.8: Rocker–ground reaction with and without the compliant connection.

6.5.3 Wheel reaction and spring-force requirement

The swingarm–wheel reaction also exhibits lower peaks in the compliant model, as shown in Figure 6.9.

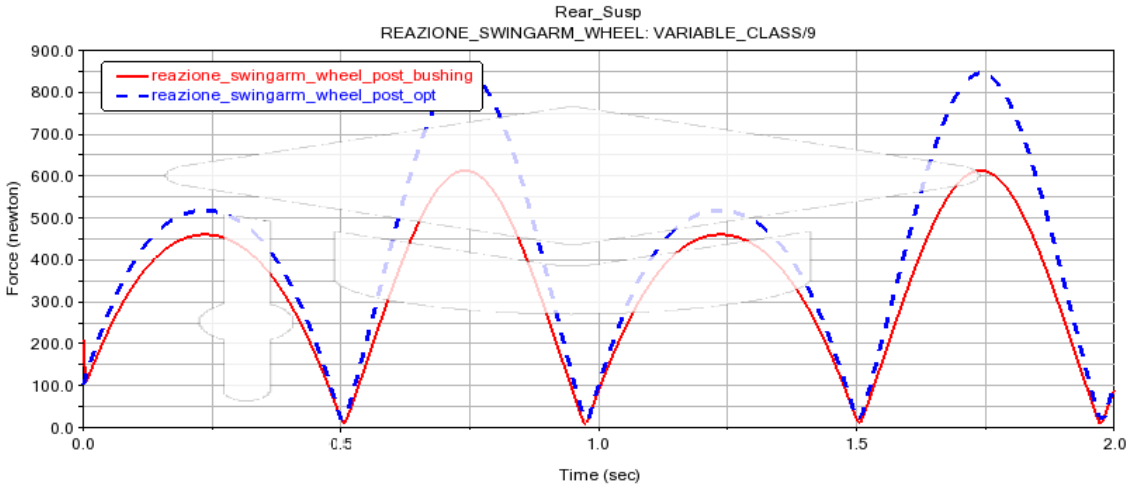


Figure 6.9: Swingarm–wheel reaction: compliant model and optimised rigid-joint reference.

The reduction in transmitted loads must be considered alongside the spring-force requirement. To check this point, the selected rigid-joint and bushing configurations are simulated again in MSC Adams Student 2025.1, using the same optimised design variables and the respective settings in Table 6.2. Both runs use the original spring-force measure, `forza_molla`, with no initial static-equilibrium calculation.

The repeated rigid-joint response is graphically consistent with the previously documented peak of 2562.8 N. With the bushing, the plotted maximum absolute force is approximately 2000 N, below the 2500 N minimum. The repeated-run values are estimates read from the plots, not new numerical objective evaluations.

Table 6.3: Spring-force assessment of the two connection models.

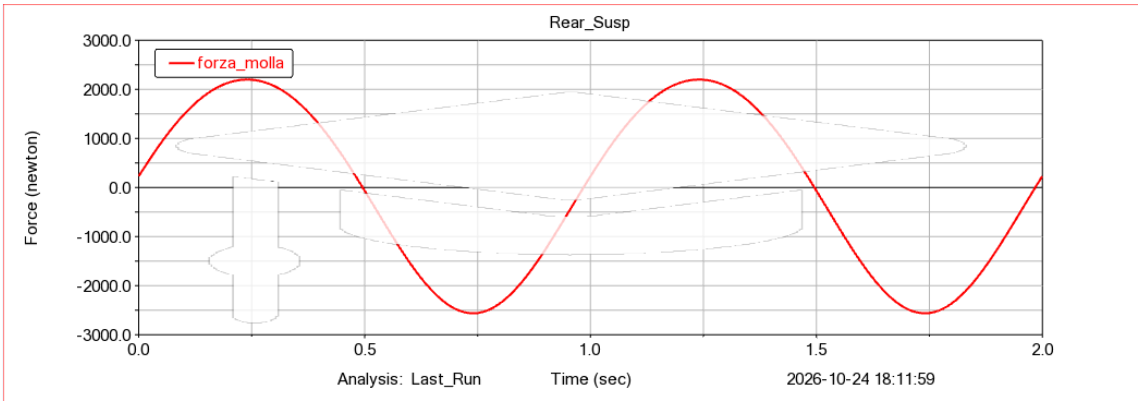
Configuration	Peak force [N]	Requirement
Optimised rigid joints	2562.8	Satisfied
Optimised geometry with bushing	≈ 2000	Not satisfied

The rigid-joint value is the original evaluated objective, supported by the repeated plot; the bushing value is a graphical estimate. The lower joint loads do not, by themselves, make the compliant configuration an acceptable replacement for the selected rigid-joint design.

6.5.4 Repeated spring-force comparison

Figure 6.10 presents the repeated simulations separately so that each response is associated with a known connection configuration. Both panels retain the original Adams output and Student Edition watermark.

(a) Optimised rigid-joint configuration



(b) Optimised geometry with bushing

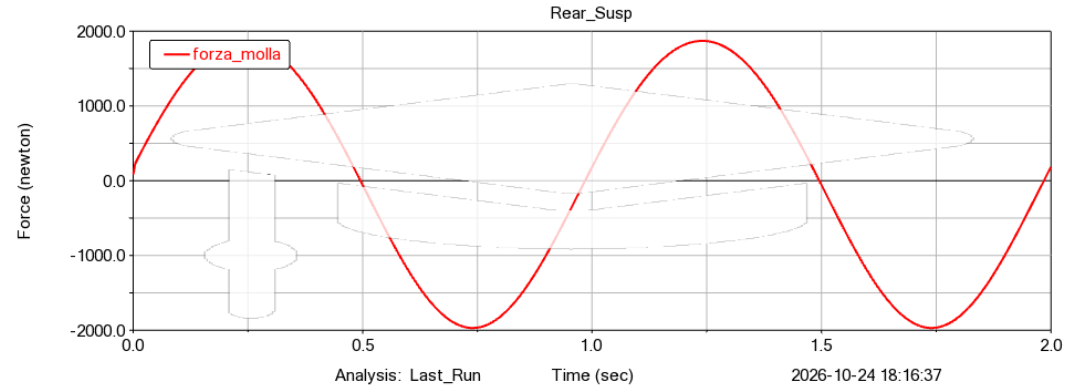


Figure 6.10: Spring–damper force in the repeated rigid-joint and compliant simulations. Each panel covers 0–2 s. The vertical scales differ: compare the numerical axis values, rather than the apparent curve heights.

The rigid-joint plot has a positive peak of approximately 2200 N and a negative peak near −2560 N. The bushing plot reaches approximately 1900 N and −2000 N. The negative peak governs the maximum absolute force in both cases.

Elastic deformation stores energy and damping dissipates energy, modifying the force response. In this study, the tested coefficients reduce the spring-force peak below the required interval. Further tuning would be needed to reconcile the lower joint reactions with the spring-force requirement; no such additional optimisation is claimed here.

6.6 Comparison of wheel kinematics

The prescribed vertical motion remains unchanged when the bushing is introduced. The displacement and velocity comparisons show overlapping responses for the compliant model and the initial reference. The same input amplitude and frequency are therefore retained throughout the study.

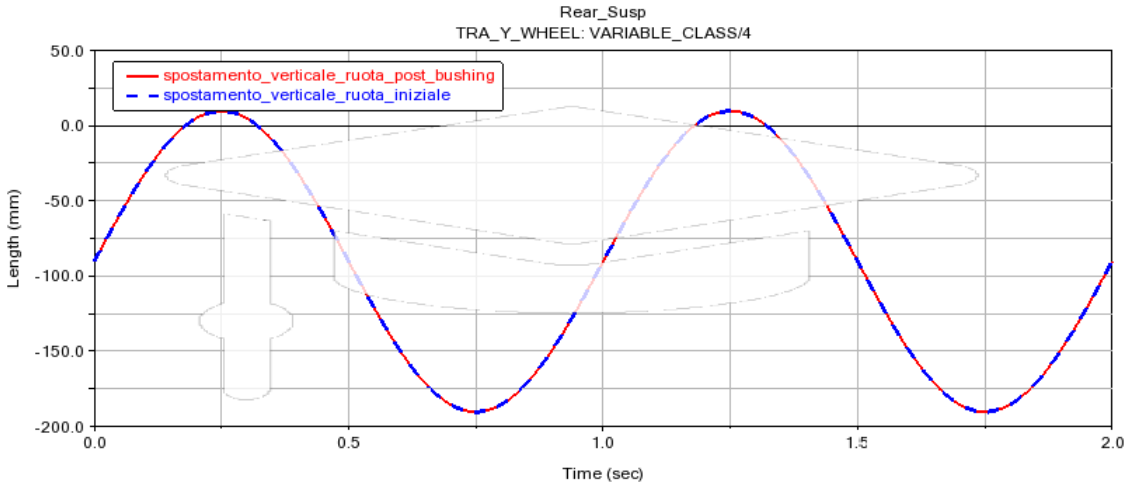


Figure 6.11: Vertical wheel displacement after introducing the bushing, compared with the initial response.

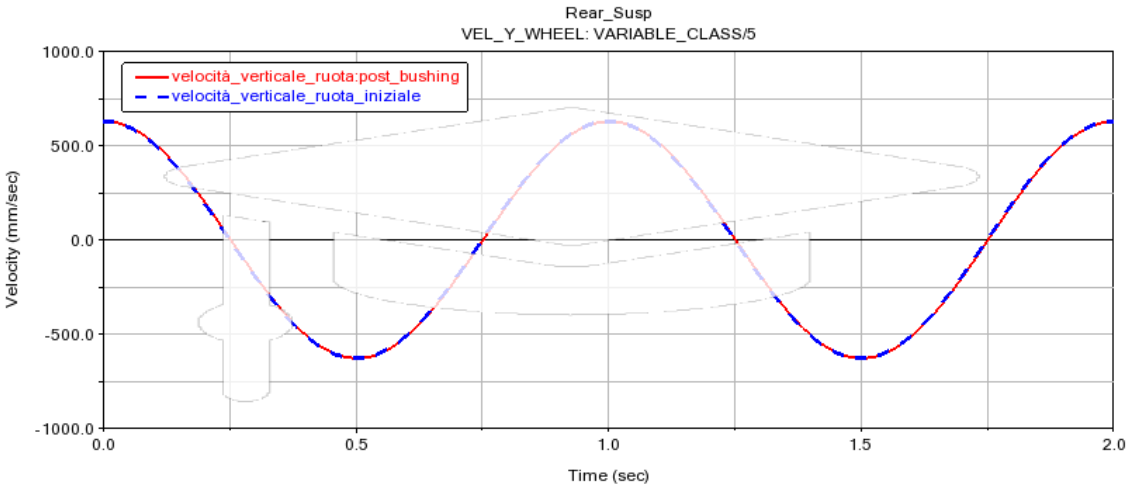


Figure 6.12: Vertical wheel velocity after introducing the bushing, compared with the initial response.

The acceleration curves also coincide, consistent with the unchanged time dependence of the prescribed input. Together, the three comparisons show that the vertical wheel kinematics are maintained while the internal load response changes.

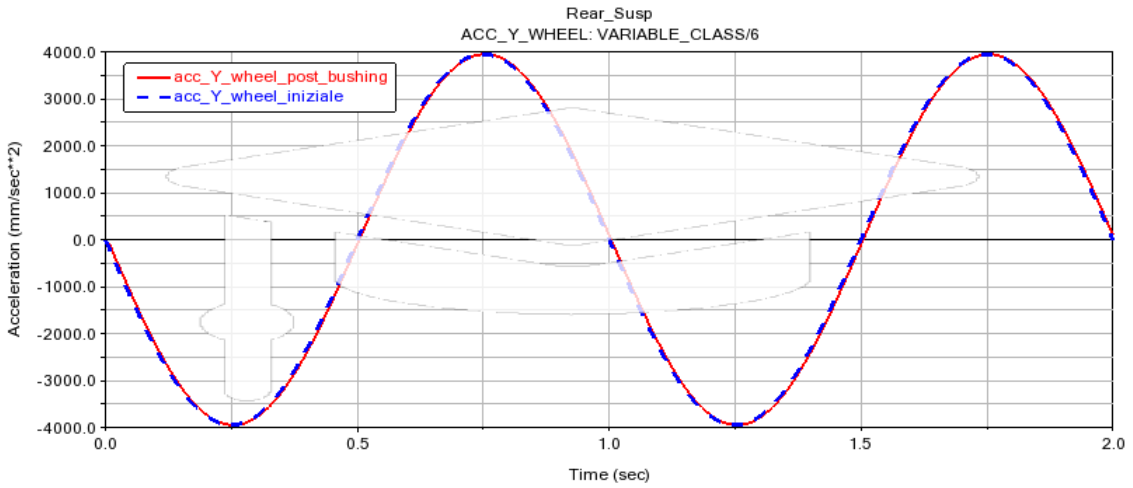


Figure 6.13: Vertical wheel acceleration after introducing the bushing, compared with the initial response.

6.7 Discussion of the compliant configuration

Replacing the cylindrical joint with a bushing introduces relative elastic motion between the connecting rod and swingarm. In the reported comparisons, the joint-reaction peaks decrease while the vertical wheel displacement, velocity and acceleration remain governed by the original sinusoidal law.

The force and kinematic results have different roles in the interpretation. The force plots show the effect of the modified connection on load transmission. The coincident vertical kinematic curves reflect the fact that the same motion remains prescribed; the input itself has not been changed to obtain the lower reactions.

The spring-force acceptance demonstrated in Chapter 5 applies to the selected rigid-joint configuration. The repeated comparison in Figure 6.10 indicates that the tested bushing configuration falls below the required interval. The compliance study therefore illustrates a load-reduction trade-off rather than a final design satisfying all the stated requirements.

The compliance study therefore extends the geometric and spring-parameter optimisation by examining the connection model. It provides a comparison between an ideal joint and a connection that can deform and dissipate energy, using the same wheel excitation.

Chapter 7

Conclusions

The project develops a working multibody model of a motorcycle rear suspension from a supplied component layout. The joints are selected and checked for redundant constraints, the wheel's vertical motion is prescribed, and a spring–damper connection is introduced between the rocker arm and ground.

The kinematic analysis reproduces the displacement, velocity and acceleration associated with a sinusoidal input with an amplitude of 100 mm and a frequency of 1 Hz. The initial force analysis identifies the connecting-rod joints as the most highly loaded connections, with a peak reaction of approximately 3670 N. The initial spring–damper peak is approximately 2982 N and lies within the specified interval.

The first optimisation reduces the main joint reaction to approximately 1750 N, but also lowers the spring–damper peak to approximately 1997 N. This configuration therefore fails the force requirement. Restoring the nominal spring parameter alone is insufficient, leading to the subsequent DOE study.

The selected DOE configuration gives a maximum beam–swingarm reaction of 2643.4 N and a peak absolute spring–damper force of 2562.8 N. Relative to the initial model, these correspond to reductions of approximately 28% and 14%, respectively. The final rigid-joint design therefore reduces the main reactions while satisfying the required spring-force interval.

The final stage replaces the beam–swingarm cylindrical joint with a bushing. The dynamic comparisons show lower joint-reaction peaks for the compliant configuration, while the imposed vertical wheel kinematics remain unchanged. The repeated spring-force comparison also shows a peak absolute force of approximately 2000 N, estimated from the plotted response. This is below the required interval, so the tested bushing configuration would require further tuning before it could satisfy the same spring-force requirement.

Overall, the study links modelling choices, force evaluation and parameter selection in a single design process. Its principal outcome is a rigid-joint suspension configuration with lower joint loading that meets the stated spring–damper requirement under the prescribed excitation. The bushing study separately illustrates the effect of connection compliance on transmitted forces. The conclusions concern this simplified model and load case; assessment of a complete motorcycle would require additional operating conditions and vehicle-level modelling.

Appendix A

Parameter reference

A.1 Design-variable record

Table A.1 records the parameter sets used in the main rigid-joint design stages. The compliant model uses the selected final geometry and spring–damper parameters, with the connection change described in Chapter 6.

Table A.1: Design variables throughout the optimisation sequence. Geometric variables are expressed in mm.

Variable	Initial	First optimisation	Selected final
DV_K	75	67.5	82.5
x_mono	30	30	70
y_mono	130	130	130
x_triangle	-30	-30	30
y_triangle	-160	-110	-110
delta_tx	10	30	30
delta_ty	5	5	15

Table A.2: Allowed intervals of the selected design variables.

Variable	Lower bound	Upper bound
DV_K	67.5	82.5
x_mono [mm]	30	70
y_mono [mm]	130	160
x_triangle [mm]	-30	30
y_triangle [mm]	-160	-110
delta_tx [mm]	10	30
delta_ty [mm]	5	15

A.2 Physical meaning of the design variables

The variable definitions below follow the supplied Adams command model. Ground coordinates describe the initial layout; the rocker attachment coordinates are expressed in the rocker's local frame.

Table A.3: Design variables and the quantities they control.

Variable	Definition
`DV_K`	Dimensionless numerical parameter linking spring stiffness, damping and preload through Equation (2.2).
`x_mono`, `y_mono`	Ground-frame X and Y coordinates of `P_cyl_ch`, the chassis attachment of the spring–damper.
`x_triangle`, `y_triangle`	Ground-frame X and Y coordinates of `P_tr_ch`, the rocker chassis pivot.
`delta_tx`, `delta_ty`	Offsets defining `P_cyl_tr`, the spring–damper attachment on the rocker, in the local coordinate expressions below.

For the rocker attachment point, the command model uses

$$x_{\text{local}} = -\delta_{tx} + L_2 \cos(G_0 - \alpha_{\text{sw}}), \tag{A.1}$$

$$y_{\text{local}} = \delta_{ty} + L_2 \sin(G_0 - \alpha_{\text{sw}}), \qquad z_{\text{local}} = 0. \tag{A.2}$$

Here L_2 is one third of `sw_l`, while G_0 and α_{sw} denote `G0` and `sw_a`, respectively. In the supplied model, `sw_l` is 520 mm, `G0` is $-10°$ and `sw_a` is $20°$. Increasing `delta_tx` therefore decreases the local X coordinate; it is not an increase in the global X position of the attachment.

The geometric variables are expressed in mm. The ground-frame Z coordinates of the two chassis attachment points are zero. These definitions distinguish changes to chassis attachment locations from changes to the spring attachment on the moving rocker.

A.3 Model environment and inertial properties

The repeated spring-force simulations use MSC Adams Student 2025.1. The command model defines mm, kg, N, s and degrees as its unit settings. Gravity acts along negative global Y with an acceleration of 9806.65 mm/s^2.

The wheel has explicitly assigned mass and inertia values, listed in Table A.4. The swingarm, connecting rod and rocker use the steel material assignment and their model geometry for mass properties; the steel density is 7.801×10^{-6} kg/mm^3. These are properties of the supplied simplified model, not measured component data.

Table A.4: Explicit wheel mass and inertia in the command model.

Property	Value	Unit
Mass	12	kg
I_{xx}	7.5×10^5	kg mm^2
I_{yy}	4.0×10^5	kg mm^2
I_{zz}	4.0×10^5	kg mm^2
I_{xy}, I_{xz}, I_{yz}	0	kg mm^2

The inertia entries refer to the wheel centre-of-mass marker used in the model. The bodies remain rigid: the steel material assignment does not introduce structural flexibility. The bushing is the compliant connection examined in Chapter 6.

The repeated runs retain the supplied motion law and spring coefficient definitions. Only the connection activation state, its associated force measures and the analysis mode/output settings are changed between the rigid and compliant checks. No new DOE or parameter optimisation is performed.

A.4 Spring–damper properties

The coupled coefficient definitions give the values in Table A.5. The intermediate nominal-parameter check uses the first optimised geometry with the nominal values in the first column.

Table A.5: Spring–damper properties in the principal configurations.

Property	Initial	First optimisation	Selected final
K [N/mm]	75	67.5	82.5
C [N s/mm]	0.750	0.675	0.825
F_0 [N]	75	67.5	82.5

A.5 Names retained in the original plots

The plots are reproduced from Adams outputs. Figure 6.10 contains the repeated spring-force checks; the other analysis plots retain the original results. Their legends retain the model's original object names so that each curve remains identifiable. The following terms provide a reading guide.

Table A.6: Terminology used in the Adams plot legends.

Original term	Meaning
`reazione`	Joint reaction force
`forza_molla`	Spring–damper force
`iniziale`	Initial configuration
`prima_opt`, `1opt`	First optimisation
`DOE`, `finale`	Selected final design
`post_bushing`	Configuration with compliant connection
`spostamento`, `tra_Y`	Vertical displacement or position output
`velocita`, `vel_Y`	Vertical velocity
`acc`, `acc_Y`	Vertical acceleration

English captions identify the bodies, quantities and configurations being compared. Force magnitudes are distinguished from signed force components and from the signed spring–damper force. Numerical values in summary tables are rounded consistently with the precision of the original results.